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The Effects of Macroeconomic Shocks: Household Financial Distress Matters

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1 Big picture

- **Question.** When a macroeconomic shock hits, how much of the consumption response is driven by **who households are**—their ex ante financial fragility—versus **what shocks they receive**?
- **Core object.** The paper studies **financial distress (FD)**, defined empirically as being at least 30 days delinquent on unsecured debt, and asks whether FD is just a proxy for harsher shocks or whether it captures deeper heterogeneity in marginal propensities to consume.
- **Setting.** The paper combines U.S. evidence from the Great Recession and the COVID recession with a quantitative life-cycle model featuring housing, mortgages, unsecured debt, bankruptcy, and delinquency.
- **Main challenge.** FD is endogenous. Households in distress have different balance sheets, different credit access, and, in the data, they also tend to get hit harder by aggregate shocks. A reduced-form comparison cannot separate these channels cleanly.
- **Main result.** Heterogeneity in household balance-sheet health matters more than heterogeneity in shock exposure. Accounting for FD changes the response of aggregate consumption, consumption inequality, and consumption-based poverty by about **25% on average**, with house-price shocks affected more than labor-income shocks.
- **Mechanism.** FD matters because it is associated with high MPC households and because the model with FD generates endogenous unsecured borrowing terms that interact with housing status. In the model, FD is not just another label; it changes incentives and equilibrium credit conditions.
- **Housing-market implication.** Allowing for FD raises housing leverage, mortgage debt, and mortgage default relative to a model that uses only a fixed ad hoc borrowing limit.
- **Why the paper matters.** It shows that balance-sheet heterogeneity should not be collapsed into a simple borrowing constraint if the researcher cares about how shocks pass through to consumption, inequality, poverty, and housing outcomes.

2 Data and measurement (key objects)

2.1 Observed financial-distress panel

- **Main FD measure.** FD is defined using the New York Fed Consumer Credit Panel / Equifax as having a credit-card account at least 30 days delinquent during the year. The paper calls this **DQ30**.
- **Alternative FD measure.** The paper also considers **CL80**, an indicator for using at least 80% of an unsecured credit line. Results are qualitatively similar, but the quantitative model focuses on DQ30 because it maps more naturally into delinquency in the model.
- **Why FD is useful empirically.** It is easy to observe, updates quickly in credit data, and does not require full balance-sheet valuation. Unlike leverage or net worth, it directly reflects that a household is close to, or already at, the edge of repayment capacity.
- **Incidence and persistence.** FD is common and persistent. Roughly 10–20% of individuals are in FD at any point in the life cycle, and current FD strongly predicts future FD.

2.2 Observed macro-shock exposure

- **Great Recession shock.** House-price changes are measured using ZIP-code-level housing data, with the paper showing that areas with higher pre-crisis FD experienced larger house-price declines.
- **COVID shock.** Labor-income and employment exposure are measured using evidence on contact-sensitive employment and survey-based earnings losses. Areas with higher prior FD suffered larger labor-market losses.
- **Empirical fact.** FD is positively correlated with later shock severity in both recessions. This is exactly why the structural model is needed.

2.3 Calibration targets and grouped economies

- **Five “economies.”** The authors sort U.S. ZIP codes into five quintiles by the incidence of FD and estimate one version of the model for each quintile. Quintile 1 is the least distressed, quintile 5 the most distressed.
- **Moments to match.** The model is disciplined by savings-to-income, homeownership, housing leverage, housing debt conditional on FD, delinquency, bankruptcy, and the persistence of FD.
- **Initial asset distribution.** Initial net financial wealth for young households is set from the Survey of Consumer Finances.
- **External evidence for validation.** The model is checked against MPC estimates from the empirical literature for both house-price and labor-income shocks.

2.4 Unobservables and timing

- **Life cycle.** A period is one year. Households enter at age 25, retire at 65, and face mortality risk until terminal age 82.
- **Preference heterogeneity.** Households differ in discount factors: a patient type with discount factor β_H and an impatient type with discount factor β_L .
- **Housing heterogeneity.** Rental-house size can differ by type, which is a reduced-form way to capture differences in the value of homeownership relative to renting.
- **Default states.** The model distinguishes repayment, informal default via delinquency, and formal default via bankruptcy.
- **Interpretation.** FD in the model is defined as being in delinquency or bankruptcy, though delinquency is quantitatively the dominant component.

3 Models and empirical specifications (all equations + interpretation)

3.1 Environment and household heterogeneity

There is a continuum of finitely lived, risk-averse households. Survival to the next period occurs with age-dependent probability ρ_t . Each household works until retirement age W and then receives retirement income.

Households differ in discount factors. A share s_L are impatient with discount factor β_L , and the rest are patient with discount factor $\beta_H \geq \beta_L$.

- **Interpretation.** This is the paper’s central source of ex ante heterogeneity. It is what ultimately generates large differences in savings behavior, FD persistence, and MPCs.
- **Why two types are enough.** The goal is not to recover a literal two-point distribution of patience, but to create an empirically tractable way to encode persistent differences in households’ willingness to borrow and repay.

3.2 Preferences and income process

Period utility over nondurable consumption c and housing services h is

$$u(c, h) = \frac{((1 - \theta)c^{1-1/\alpha} + \theta h^{1-1/\alpha})^{(1-\gamma)/(1-1/\alpha)}}{1 - \gamma}.$$

- γ is risk aversion.
- α governs intratemporal substitution between housing and nondurables.
- θ controls the housing expenditure share.

Working-age income follows a restricted-income-profile process:

$$\log(y_{n,t}^i) = \ell(n) + z_{n,t}^i + \varepsilon_{n,t}^i,$$

with persistent component

$$z_{n,t}^i = z_{n,t-1}^i + e_{n,t}^i.$$

- **Interpretation.** The model has standard life-cycle earnings risk, but the income process itself does not vary by FD quintile. Differences across quintiles come primarily from preference composition and rental-house-size heterogeneity, not from directly imposing different idiosyncratic income processes.

3.3 Assets, housing, and the meaning of FD

Households choose nondurable consumption, housing, mortgages, and unsecured financial assets:

$$(c, h, m', a').$$

If $a' > 0$, the household saves at the risk-free rate r . If $a' < 0$, the household borrows unsecured and faces a price schedule that depends on future repayment risk.

Households can be:

- renters,
- homebuyers,
- existing homeowners.

They may also be in one of three debt states:

- **paying**,
- **delinquent** (informal default),
- **bankrupt** (formal default).

Delinquency is modeled as skipped repayment on unsecured debt. Debt is discharged with probability η ; otherwise it rolls over at penalty rate r^R . Bankruptcy wipes out unsecured debt but imposes filing cost f .

- **Interpretation.** This is the paper's key modeling move. FD is not approximated by a borrowing constraint; it is embedded as an endogenous repayment state.
- **Why delinquency matters.** Informal default is empirically common and highly persistent, so it gives the model a much closer connection to observed distress than bankruptcy-only environments.

3.4 Nonhomeowners: renting, borrowing, and buying

A renter of type j with current assets a and income y who pays unsecured debt chooses next period assets a' subject to

$$c + q_{j,n}^R(a', y)a' = y + a.$$

If the renter files for bankruptcy,

$$c = y - f.$$

If the renter becomes delinquent, current consumption is simply

$$c = y,$$

and next period unsecured assets satisfy

$$a' = \begin{cases} 0, & \text{with probability } \eta, \\ (1 + r^R)a, & \text{with probability } 1 - \eta. \end{cases}$$

If a nonhomeowner buys a house of size h' , chooses mortgage m' , and unsecured assets a' , the budget set is

$$c + q_{j,n}^a(h', m', a', y)a' = y + a + q_{j,n}^m(h', m', a', y)m' - \mathbf{1}_{m' > 0}\xi_M - (1 + \xi_B)ph',$$

subject to the loan-to-value constraint

$$q_{j,n}^m(h', m', a', y)m' \leq \lambda ph'.$$

- **Interpretation.** A homebuyer's feasible set depends on endogenous mortgage pricing and endogenous unsecured debt pricing. The two credit markets interact.
- **Economic meaning.** Buying a house is not only a consumption choice but also a financing choice, because homeownership changes the household's outside options in distress states.

3.5 Homeowners: payment, refinancing, foreclosure, and sale

An existing homeowner who makes the mortgage payment and also pays unsecured debt faces

$$c + q_{j,n}^a(h, m(1 - \delta), a', y)a' = y + a - m.$$

If the homeowner refinances the mortgage, then current mortgage obligations are prepaid. Let the mortgage prepayment value with n periods remaining be

$$q_n^* = \frac{1 - \left(\frac{1-\delta}{1+r}\right)^{n+1}}{1 - \frac{1-\delta}{1+r}}.$$

The refinancing budget constraint becomes

$$c + q_{j,n}^a(h', m', a', y)a' = y + a - q_n^*m + q_{j,n}^m(h', m', a', y)m' - \mathbf{1}_{m' > 0}\xi_M.$$

If the owner sells the current house and becomes a renter,

$$c + q_{j,n}^R(a', y)a' = y + a + ph(1 - \xi_S) - q_n^*m.$$

If the owner sells and buys another house,

$$\begin{aligned} c + q_{j,n}^a(h', m', a', y)a' &= y + a + ph(1 - \xi_S) - q_n^*m \\ &\quad + q_{j,n}^m(h', m', a', y)m' - \mathbf{1}_{m' > 0}\xi_M - (1 + \xi_B)ph', \end{aligned}$$

again subject to the LTV restriction.

Mortgage default sends the household into renter status, after which it chooses whether to pay, become delinquent, or file for bankruptcy on unsecured debt.

- **Interpretation.** Figure 6 in the paper is important because it makes clear that homeowners have many more adjustment margins than renters.
- **Key idea.** Homeowners can refinance, extract equity, sell, or default on the mortgage. These extra options directly affect how unsecured lenders price their debt.

3.6 Debt prices

Mortgage prices are set by risk-neutral lenders making zero expected discounted profit. Mortgage pricing can be written as

$$q_{j,n}^m(h', m', a', y) = \frac{q_{j,n}^{m,\text{pay}} + q_{j,n}^{m,\text{prepay}} + q_{j,n}^{m,\text{default}}}{1 + r}.$$

Unsecured debt pricing for a household of type j can be written as

$$q_{j,n}^a(h', m', a', y) = \frac{q_{j,n}^{a,\text{pay}} + q_{j,n}^{a,\text{DQ}}}{1 + r}.$$

- **Interpretation.** Mortgage prices depend on whether the loan is repaid, prepaid, or defaulted. Unsecured debt prices depend on whether the household repays or becomes delinquent.
- **Crucial interaction.** For homeowners, unsecured lenders know the borrower may refinance, extract equity, sell the house, or default on the mortgage rather than on unsecured debt. This improves repayment prospects on unsecured borrowing.
- **Bottom line.** Endogenous credit pricing is the channel through which FD affects housing choices and through which housing status feeds back into unsecured borrowing conditions.

3.7 Externally set parameters and calibration choices

A period is one year. The paper sets:

$$\begin{aligned} W = 65, \quad r = 0.03, \quad \gamma = 2, \quad \alpha = 0.5, \quad \theta = 0.14, \\ \lambda = 0.9, \quad \beta_H = 1.00, \quad \beta_L = 0.80, \quad \eta = 0.25, \quad r^R = 0.20, \end{aligned}$$

along with bankruptcy cost, transaction costs, mortality, retirement-income parameters, and income-shock variances from external sources or standard calibrations.

- **House price normalization.** House prices are set at $p = 3.33$ to line up with home-value-to-income ratios.
- **Rental-house-size asymmetry.** Patient types are assigned a very small rental-house size, so homeownership is especially valuable to them; impatient types have rental-house size h_L^R estimated from the data.

3.8 Estimating the remaining heterogeneity

The key estimated parameters, for each FD quintile economy, are:

- the share of impatient types, s_L ,
- the rental-house size for impatient types, h_L^R .

The model matches moments on:

- savings-to-income,
- homeownership,
- housing leverage,
- housing debt conditional on FD,
- delinquency,
- bankruptcy,

- persistence of FD.

The estimates imply:

$$s_L : 0.29 \rightarrow 0.58 \quad \text{from Q1 to Q5,}$$

and

$$h_L^R : 4.61 \rightarrow 2.95 \quad \text{from Q1 to Q5.}$$

- **Interpretation.** More distressed places contain a much larger share of impatient households.
- **Economic meaning of h_L^R .** The lower value of h_L^R in more distressed quintiles helps the model reproduce the fact that the homeownership gap between distressed and average households is smaller there.

3.9 Aggregate shock calibration

The paper then hits each quintile economy with shocks that mimic the observed FD–shock correlation. For house prices, the calibrated average declines are:

$$6.9, 8.6, 10.1, 10.9, 11.4$$

percent from Q1 to Q5.

For the labor-income shock, the share suffering a severe 50% earnings loss rises from

$$14.4\% \text{ in Q1 to } 21.7\% \text{ in Q5,}$$

and average earnings loss rises from

$$8.5\% \text{ to } 12.3\%.$$

- **Interpretation.** High-FD households are both more fragile and more exposed.
- **Why this matters.** The paper can then separately ask whether the aggregate results are coming from different people or from different shocks.

3.10 Model validation through MPCs

The model’s aggregate MPCs line up with external evidence:

$$\text{MPC}_{\text{house, all}} = 0.071, \quad \text{MPC}_{\text{house, owners}} = 0.088, \quad \text{MPC}_{\text{income}} = 0.299.$$

Across FD quintiles, the labor-income MPC rises from

$$0.242 \text{ in Q1 to } 0.384 \text{ in Q5.}$$

Among ages 30–50, owners in FD have an MPC out of house-price shocks of 0.111, versus 0.047 for owners not in FD. For income shocks, households in FD have an MPC of 0.689 versus 0.268 for those not in FD.

- **Interpretation.** FD is a good empirical tag for high-MPC households.
- **Deeper point.** Sorting on observed FD produces very similar rankings to sorting on the model’s unobserved preference type.

4 Identification and instruments (what makes the estimates credible)

4.1 What fails in a naive interpretation

- A simple correlation between FD and later consumption responses is not informative enough because households in FD differ in both their underlying characteristics and the shocks they receive.
- A standard incomplete-market model with a fixed borrowing limit is too crude because it cannot reproduce the interaction between unsecured debt, homeownership, refinancing, and mortgage default.

4.2 What identifies household heterogeneity

- **Discount-factor heterogeneity** is identified by moments on savings-to-income, the level of delinquency, and the persistence of delinquency and bankruptcy.
- **Rental-house-size heterogeneity** is identified by matching homeownership rates, bankruptcy, and especially the gap between homeownership for distressed households and the overall homeownership rate.
- The combination of both is essential. Discount-factor heterogeneity alone gets the MPC patterns roughly right, but it badly misses the homeownership gap for distressed households.

4.3 Why the restricted-model comparisons matter

The paper estimates two restricted models:

- only discount-factor heterogeneity,
- only rental-house-size heterogeneity.

These comparisons show:

- discount-factor heterogeneity is the main driver of the relationship between FD and MPCs;
- rental-house-size heterogeneity is important for matching the housing-status facts;
- both are needed to match the full joint distribution of FD, debt, and homeownership.

4.4 What identifies the shock-exposure channel

- The model first matches the observed correlation between FD and shock intensity.
- Then it shuts that correlation down by imposing common shocks across FD quintiles while holding the household heterogeneity fixed.
- This counterfactual isolates the contribution of **shock exposure** from the contribution of **FD itself**.

4.5 Why the model is believable quantitatively

- Aggregate and group-specific MPCs line up with the empirical literature on house-price and income shocks.
- The estimated share of impatient types rises smoothly across FD quintiles.
- The model reproduces the persistence of FD and the strong homeownership gap for distressed households.

4.6 Relation to the literature

- Relative to household-finance models with only ad hoc borrowing constraints, the paper embeds delinquency and bankruptcy directly into equilibrium debt pricing.
- Relative to the bankruptcy literature, it uses FD not only as an adjustment margin but also as a source of ex ante heterogeneity.
- Relative to empirical work on heterogeneous MPCs, it provides a structural interpretation: FD is informative because it tracks deeper differences in effective impatience and credit access.

5 Tables and figures

5.1 Figure 1: FD is common and persistent

Read:

- About 10–20% of people are in FD at a given age.
- FD is very persistent: someone in FD today is much more likely to still be in FD two, four, or even ten years later.
- This is why FD is not just temporary bad luck; it is a meaningful state variable for the model.

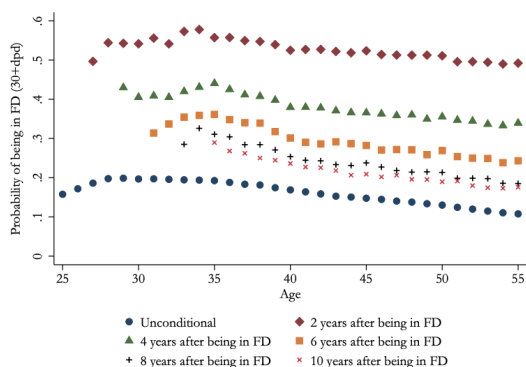


Figure 1
The incidence of persistence of FD over the life cycle
This figure plots the average probability of being in FD, defined as an individual having a credit card account 30 days or more delinquent at some point during the year. Source: FRBNY Consumer Credit Panel/Equifax.

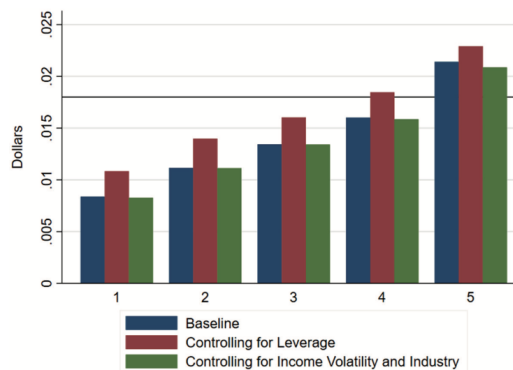


Figure 2
Marginal propensity to consume out of a dollar change in home prices by quintile of DQ30 in 2002
Group means are weighted by the number of owner-occupied housing units per county as of 2006. The horizontal line corresponds to the mean MPC out of autos estimated at the ZIP code level by Mian, Rao, and Sufi (2013) in the fifth column of their table V. Sources: IRS Survey of Income, FRBNY Consumer Credit Panel/Equifax, Census Bureau, Zillow, Survey of Consumer Finances.

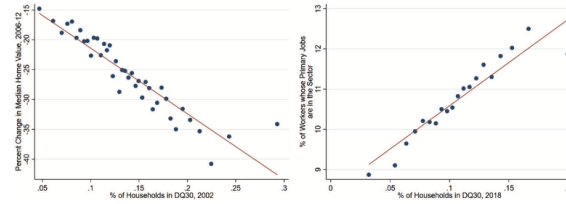


Figure 3
The correlation between FD and house-price shocks during GR (left) and contact-sensitive employment (right)
 The panel on the left includes ZIP Codes for which we also have Core Logic data. FD is measured as DQ30, which is the share of individuals that are at least 30 days delinquent on a credit card at some point in a given year. For ease of viewing, the data have been divided into 40 bins with respect to DQ30, and each dot represents the mean of that bin. In the left panel, each bin is weighted by the housing wealth in each ZIP code in that bin as of 2006. In right panel, each bin is weighted by the number of households in each ZIP code included in the bin. *Sources:* Zillow, Census, LEHD LODES, and FRBNY Consumer Credit Panel/Equifax.

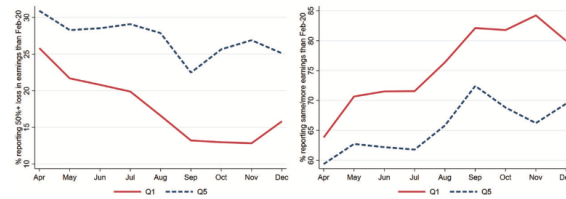


Figure 4
Earnings changes in 2020 by quintile of FD: Losses of at least 50% (left) and no change (right)
Sources: Bick and Blandin (2021) and FRBNY Consumer Credit Panel/Equifax.

5.2 Figure 2: ZIP codes with more FD have larger MPCs out of housing shocks

Read:

- The MPC out of house-price shocks rises from below 1 cent to above 2 cents across FD quintiles in the reduced-form evidence.
- This relationship survives controls for leverage, income volatility, and industry composition.
- So FD captures something beyond standard vulnerability measures.

5.3 Figures 3 and 4: High-FD places were hit harder in both recessions

Read:

- During the Great Recession, higher-FD ZIP codes had larger house-price collapses.
- Before and during COVID, higher-FD places had more contact-sensitive employment and larger earnings losses.
- Therefore, FD is correlated with shock exposure in the data, making the model's decomposition essential.

5.4 Figures 5 and 6: The model's choice structure

Read:

- Figure 5 lays out the renter's options: pay, go delinquent, file bankruptcy, or buy a home.

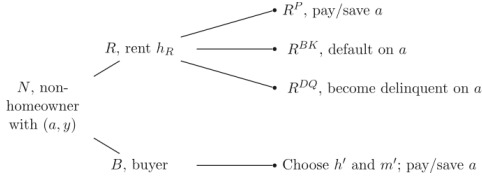


Figure 5
Decision tree of a nonhomeowner

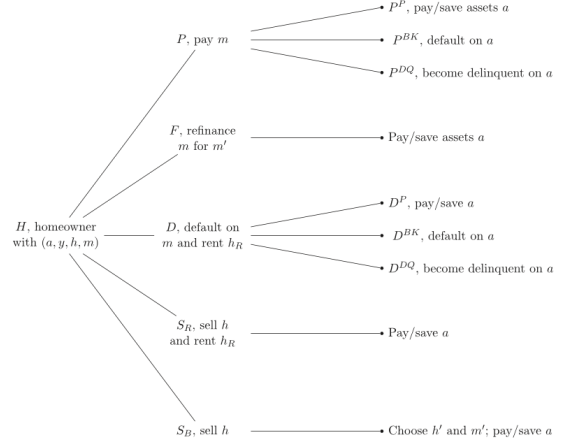


Figure 6
Decision tree of a homeowner

- Figure 6 shows that homeowners have many more margins of adjustment: keep paying, refinance, default on the mortgage, sell and rent, or sell and buy again.
- These extra homeowner options are why unsecured lenders treat homeowners differently from renters.

5.5 Table 1: Externally set parameters

Read:

- The calibration is disciplined by standard values for the life cycle, income process, utility curvature, housing costs, bankruptcy costs, and retirement income.
- The paper fixes $\beta_H = 1.00$ and $\beta_L = 0.80$, uses an LTV cap of 0.9, and calibrates delinquent-debt rollover and discharge to match unsecured-credit facts.
- The estimated part of the model is therefore focused tightly on the cross-quintile composition of households.

5.6 Tables 2 and 3: Model fit and heterogeneity across FD quintiles

Read:

- The model matches the broad cross-quintile patterns in savings, homeownership, leverage, delinquency, bankruptcy, and FD persistence.
- The share of impatient households rises strongly with FD, from 0.29 in Q1 to 0.58 in Q5.
- The estimated rental-house size for impatient types falls from 4.61 to 2.95 across quintiles.
- The paper's interpretation is that more distressed places contain more effectively impatient households and different homeownership incentives.

Table 1
Externally set parameters

Parameter	Value	Definition	Basis
l	—	Life-cycle component of income	Kaplan and Violante 2010
W	65	Retirement age	U.S. Social Security
ρ_n	—	Mortality age profile	Kaplan and Violante 2010
α_2	—	Initial net financial asset distribution	SCF 1998–2016
σ_ϵ^2	0.05	Variance of ϵ	Kaplan and Violante 2010
σ_e^2	0.01	Variance of e	Kaplan and Violante 2010
r	0.03	Risk-free rate	Standard
γ	2	Risk aversion	Standard
α	0.5	Elasticity of substitution	Standard
θ	0.11	Consumption weight of housing	Hatchondo, Martinez, and Sánchez 2015
ζ_B	0.03	Cost of buying a house, households	Gruber and Martin 2003
ζ_S	0.03	Cost of selling a house, households	Gruber and Martin 2003
$\zeta_{\bar{S}}$	0.22	Cost of selling a house, banks	Pennington-Cross 2006
ζ_M	0.15	Cost of signing a mortgage	U.S. Federal Reserve
δ	0.02	Mortgage payment decay	Average inflation
A_0	0.7156	Replacement ratio	U.S. Social Security
A_1	0.04	Replacement ratio	U.S. Social Security
A_2	0.14	Replacement ratio	U.S. Social Security
λ	0.9	LTV limit	Positive down payment
f	0.028	Cost of filing for bankruptcy/mean(inc)	Livshits, MacGee, and Tertilt 2007
r_R	0.2	Rollover rate on delinquent debt	Livshits, MacGee, and Tertilt 2007
β_H	1.00	Discount factor of patient types	Athreya, Mustre-del Río, and Sánchez 2019
β_L	0.80	Discount factor of impatient types	Athreya, Mustre-del Río, and Sánchez 2019
h_H^R	0.001	Size of rental house for patient types	See text
p	3.33	House prices	See text
η	0.25	Discharge rate of delinquent debt	See text

Table 2
Model fit by quintile of FD

	Q1		Q2		Q3		Q4		Q5	
	Data	Model	Data	Model	Data	Model	Data	Model	Data	Model
Savings/inc	2.44	1.73	1.96	1.53	1.78	1.35	1.57	1.23	1.06	1.03
Home ownership*	73.42	76.63	68.35	71.04	64.14	62.34	59.88	61.63	57.03	52.58
Housing leverage	44.45	30.06	46.60	36.56	46.17	40.89	47.69	44.34	45.32	44.74
Housing debt > 0*	49.77	28.75	44.67	28.72	39.83	25.04	36.29	27.43	31.86	24.2
Housing debt > 0* conditional on FD	33.31	32.18	30.73	30.75	28.37	20.89	26.9	27.44	25.99	21.37
Housing debt/inc	1.53	0.86	1.65	1.03	1.68	1.17	1.70	1.31	1.60	1.52
Mortg def rate*	1.52	1.3	1.81	1.69	2.24	2.2	2.58	2.18	3.33	2.49
DQ rate*	8.98	10.22	12.65	13.4	15.43	16.22	18.27	18.55	23.93	22.3
BK rate*	0.39	0.59	0.55	0.67	0.63	0.65	0.65	0.69	0.64	0.66
Persistence of FD:										
Over 2 yr	9.2	6.5	8.05	5.31	6.82	4.87	5.89	4.03	4.83	3.58
Over 4 yr	6.15	5.3	5.36	4.21	4.57	3.74	3.99	3.17	3.2	2.77
Over 5 yr	5.36	5.31	4.63	4.18	3.98	3.61	3.45	3.09	2.86	2.68
Over 6 yr	4.86	5.29	4.17	4.15	3.57	3.58	3.2	3.07	2.58	2.64
Over 8 yr	3.89	5.13	3.56	4.11	2.95	3.52	2.61	3.01	2.19	2.59
Over 10 yr	3.4	4.38	3.0	3.68	2.66	3.23	2.37	2.86	2.05	2.47
SSE	1.15		0.67		0.58		0.42		0.33	

The quintiles are based on the distribution of FD, with Q1 being the least distressed quintile, and Q5 the most. * in percent. SSE is the sum of squared errors. "Savings/Income" represents mean net financial wealth divided by mean income.

Table 3
Parameter estimates by quintile of FD

Parameter	Q1	Q2	Q3	Q4	Q5
s_L	0.29 (0.09)	0.37 (0.05)	0.45 (0.04)	0.50 (0.05)	0.58 (0.03)
h_L^R	4.61 (0.07)	3.77 (0.02)	3.92 (0.06)	2.99 (0.01)	2.95 (0.01)

The quintiles are based on the distribution of FD, with Q1 being the least distressed quintile and Q5 the most. Asymptotic standard errors appear in parentheses.

5.7 Tables 4 and 5: Shock calibration and model validation

Table 4
Calibrated house-price and labor-income shocks by quintile of FD

	Average decline in house prices	Percent of population with earnings loss of:			Average earnings loss
		0%	25%	50%	
Q1	6.9	80.3	5.3	14.4	8.5
Q2	8.6	79.3	5.6	15.1	8.9
Q3	10.1	78.2	5.1	16.7	9.6
Q4	10.9	76.5	5.9	17.6	10.3
Q5	11.4	72.4	5.9	21.7	12.3

The quintiles are based on the distribution of FD, with Q1 being the least distressed quintile, and Q5 the most. Statistics for each quintile were computed using data from Bick and Blandin (2021) for labor income and Zillow for house prices.

Table 5
Model-implied MPCs

	Aggregate	Q1	Q2	Q3	Q4	Q5
House-price shocks, all	0.071	0.065	0.074	0.067	0.076	0.078
House-price shocks, owners	0.088	0.083	0.086	0.086	0.095	0.095
Labor-income shocks	0.299	0.242	0.274	0.314	0.334	0.384

This table presents marginal propensities to consume (in dollars) out of permanent house-price shocks and transitory income shocks. In the case of labor-income shocks, the calculation is restricted to working-age individuals. The quintiles are based on the distribution of FD, with Q1 being the least distressed quintile, and Q5 the most.

Read:

- Shock exposure worsens with FD: house-price declines and earnings losses are both larger in high-FD quintiles.
- The model-implied aggregate MPCs are realistic: about 7 cents for house-price shocks and 30 cents for transitory income shocks.
- The distribution of MPCs is steep across FD quintiles, especially for labor-income shocks.

5.8 Tables 6 and 7: Why FD works as a tag for high-MPC households

Read:

- Discount-factor heterogeneity is the main reason MPCs increase with FD.
- Among ages 30–50, owners in FD have an MPC out of house-price shocks of 11 cents, versus under 5 cents for owners not in FD.
- For income shocks, individuals in FD have an MPC near 69 cents, versus 27 cents outside FD.

Table 6
Model-implied MPCs by heterogeneity specification

	Aggregate	Q1	Q2	Q3	Q4	Q5
House-price shocks, all						
Baseline model	0.071	0.065	0.074	0.067	0.076	0.078
β -het model	0.073	0.064	0.071	0.074	0.077	0.085
h^R -het model	0.076	0.080	0.074	0.079	0.092	0.047
House-price shocks, owners						
Baseline model	0.088	0.083	0.086	0.086	0.095	0.095
β -het model	0.078	0.059	0.068	0.078	0.092	0.107
h^R -het model	0.084	0.090	0.077	0.091	0.102	0.051
Labor-income shocks						
Baseline model	0.299	0.242	0.274	0.314	0.334	0.384
β -het model	0.289	0.246	0.271	0.297	0.314	0.357
h^R -het model	0.270	0.266	0.271	0.265	0.276	0.276

See Table 5. The only β -het rows present the results of a restricted version of the baseline model where individuals only differ in their discount factors and are offered rental houses of the same size. The only h^R -het rows present the results of a restricted version of the baseline model where individuals only differ in the size of rental houses offered, but their discount factors are the same.

Table 7
Model-implied MPCs by FD and heterogeneity type

	All	In FD	No FD	Type- L	Type- H
House-price shocks, all	0.034	-0.013	0.041	0.139	-0.008
House-price shocks, owners	0.057	0.111	0.047	0.189	-0.007
Labor-income shocks	0.331	0.689	0.268	0.672	0.090

This table presents marginal propensities to consume (in dollars) out of permanent house-price shocks and transitory income shocks for individuals ages 30 to 50. The columns labeled "in FD" and "no FD" refer to individuals who, in the steady state (i.e., absent the corresponding shock), are in FD or not, respectively.

- These differences are very close to the model's differences between impatient and patient types, so FD is an observable proxy for the unobservable type.

5.9 Table 8: Main quantitative result—FD changes how shocks pass through

Read:

- For house-price shocks, the baseline model predicts a 1.8% fall in aggregate consumption, a 4.85% decline in the consumption p90/p10 ratio, and a 3.06% decline in consumption poverty.
- Relative to the no-FD model, FD amplifies the response to house-price shocks by about 30% on average.
- For labor-income shocks, FD attenuates the rise in inequality and poverty but amplifies the fall in aggregate consumption; the average effect is about 19%.
- This is the paper's headline result: FD changes macro transmission materially.

Table 8
The amplification role of FD by shock and consumption measure

	Baseline model (1)	Alternative model (2)	Amplification (3)	Amplification (reestimated) (4)
House-price shocks				
Consumption p90/p10	-4.85	-3.92	19.25	50.31
Consumption-based poverty	-3.06	-1.62	47.09	44.56
Aggregate consumption	-1.80	-1.36	24.50	10.35
Average absolute value			30.28	35.07
Labor-income shocks				
Consumption p90/p10	14.63	16.22	-10.89	-5.34
Consumption-based poverty	16.39	21.33	-30.17	-45.61
Aggregate consumption	-3.31	-2.78	15.85	8.26
Average absolute value			18.97	19.73

Columns 1 and 2 are measured as percent deviations from steady state. In the housing shock case, these are average changes over three periods following the shock. In the labor-income shock case, the change is measured only in the period of the shock and is calculated over the working-age population. Columns 3 and 4 measure amplification as a percentage of the corresponding value in column 1. In column 3, the amplification is calculated between the baseline model (1) and the alternative model (2). In column 4, the amplification is calculated between the baseline model (1) and the reestimated alternative model (unreported). Details of the alternative and reestimated alternative models appear in the [Internet Appendix D](#). The average absolute value represents the average of the absolute values of either column 3 or 4.

Table 9
The importance of the correlation of FD with aggregate shocks by shock and consumption Measure

	Correlation of FD w/shocks (1)	FD alone (2)	FD alone (reestimated) (3)
House-price shocks			
Consumption p90/p10	5.09	14.16	45.22
Consumption-based poverty	-10.96	58.04	55.52
Aggregate consumption	1.89	22.60	8.45
Average absolute value		31.60	36.40
Labor-income shocks			
Consumption p90/p10	9.10	-19.99	-14.44
Consumption-based poverty	6.52	-36.68	-52.12
Aggregate consumption	1.51	14.34	6.75
Average absolute value		23.67	24.44

Column 1 presents the percent difference in the corresponding measure between the baseline model with correlated (column 1 from Table 8) and uncorrelated shocks, as a fraction of the former. Column 2 presents the percent difference in the corresponding measure between the baseline model with uncorrelated shocks and the alternative model with no FD (column 2 from Table 8) as a fraction of the former. Column 3 presents the percent difference in the corresponding measure between the baseline model with uncorrelated shocks and the reestimated alternative model with no FD as a fraction of the former. The average absolute value represents the average of the absolute values of either column 2 or 3.

5.10 Table 9: The FD—shock correlation matters less than FD itself

Read:

- When the paper removes the correlation between FD and aggregate shock size, the main conclusions barely move.
- For house-price shocks, FD alone accounts for essentially all of the amplification effect.

Table 10
The housing market implications of FD

	Q1		Q2		Q3		Q4		Q5	
	Baseline model	Alt. model	Baseline model	Alt. model	Baseline model	Alt. model	Baseline model	Alt. model	Baseline model	Alt. model
Home ownership*	76.63	70.54	71.04	64.56	62.34	56.43	61.63	55.08	52.58	46.10
Housing leverage	30.06	15.61	36.56	19.00	40.89	18.76	44.34	22.70	44.74	22.21
Housing debt > 0*	28.75	21.82	28.72	21.46	25.04	17.01	27.43	19.72	24.2	15.96
Housing debt/Inc	0.86	0.54	1.03	0.64	1.17	0.64	1.31	0.80	1.52	0.91
Mortg def rate (%)	1.3	1.08	1.69	0.66	2.2	1.13	2.18	0.76	2.49	0.86

- For labor-income shocks, the correlation contributes slightly more, but still modestly.
- So the dominant force is different households responding differently to the same shock, not merely different households getting different shocks.

5.11 Table 10: FD changes the housing market too

Read:

- With FD, homeownership is higher, especially among risky types.
- Housing leverage is roughly twice as high as in the no-FD alternative model.
- Mortgage debt-to-income ratios and mortgage default rates are much higher with FD.
- This happens because unsecured lenders give homeowners better borrowing terms when home equity extraction, sale, and mortgage default are available as outside options.

6 Short summary

- **Conclusion.** The paper shows that household financial distress is quantitatively important for the transmission of macroeconomic shocks into consumption.
- **Identification insight.** The key distinction is between heterogeneity in households and heterogeneity in shock exposure, and the model shows the former is more important.
- **Empirical lesson.** FD is an observable and useful measure for identifying high-MPC, financially fragile households, and modeling it explicitly changes both consumption and housing-market predictions.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- A model that includes delinquency and bankruptcy as part of household credit choice delivers materially different predictions from a model that only imposes a fixed borrowing constraint.
- The largest part of the action comes from ex ante household heterogeneity encoded by FD, not from the fact that distressed households happened to receive worse shocks in the Great Recession or COVID.
- Therefore, balance-sheet heterogeneity is central for understanding not just aggregate consumption, but also consumption inequality, consumption poverty, and mortgage-market outcomes.

7.2 Economic intuition

- Financial distress is informative because it tells us who is close to the margin of repayment and therefore who has both high MPCs and fragile credit access.
- Once FD is modeled explicitly, credit terms become endogenous. Homeownership affects unsecured borrowing conditions because houses can be refinanced, sold, or strategically defaulted upon.
- This creates a rich interaction between consumer finance and housing finance that a one-line borrowing constraint cannot capture.

7.3 Takeaway

This paper is best understood as a warning against treating household heterogeneity as a detail that can be folded into a representative-agent or fixed-borrowing-limit model. The paper's main message is that **financial distress matters in its own right**. It matters because it identifies the households with high MPCs, because it changes how borrowing is priced, and because it alters the composition of homeowners and defaulters. Once these channels are recognized, the aggregate effects of macro shocks look meaningfully different. For work on household balance sheets, housing, credit policy, recession transmission, or inequality, this paper is a very useful benchmark.